

FastAPI Question & Answers

1. What is CRUD in Fast API?

CRUD stands for Create, Read, Update, and Delete operations. It represents the basic operations used to manage data in an application. FastAPI implements CRUD using HTTP methods like POST, GET, PUT/PATCH, and DELETE.

Example: Creating a student using POST /students.

2. Why do we use the POST method for inserting data?

POST is used to send data to the server to create a new resource. The data is sent in the request body, making it secure and not visible in the URL. It is commonly used for adding new records.

Example: POST /students with { "name": "Amit", "age": 20 }.

3. Difference between PUT and PATCH

PUT updates or replaces the entire resource with new data. PATCH updates only specific fields of a resource without changing the whole data. PUT is full update, PATCH is partial update.

Example: PUT /students/1 replaces full data, PATCH /students/1 updates only age.

4. What is the path parameter?

A path parameter is a dynamic value in the URL used to identify a specific resource, like /students/{id}. It helps retrieve or modify a particular record. It is passed directly in the route.

Example: /students/5 fetches student with ID 5

5. What is the request body in Fast API?

The request body is the data sent from the client to the server in JSON format. It is used in POST, PUT, and PATCH requests. FastAPI reads it and converts it into Python objects using Pydantic models.

Example: { "name": "Rahul", "age": 22 } sent in POST request.

6. Why do we use Pydantic models?

Pydantic models are used for data validation and structure enforcement. They ensure that incoming data follows the correct type and format. This helps reduce errors and improves API reliability.

Example: Student(name: str, age: int) validates input automatically.

7. What status code is returned for successful deletion?

A successful deletion typically returns status code **204 No Content**. It means the resource was deleted successfully and there is no response body. Sometimes 200 OK is also used.

Example: DELETE /students/1 → 204 No Content.

8. How can we check whether a student exists before updating?

We first search the student using the given ID in the database or list. If the student exists, we proceed with the update. If not, we return a 404 Not Found error.

Example: Check /students/3 before applying update.

9. What happens if duplicate IDs are inserted?

If duplicate IDs are inserted, it can cause data inconsistency or overwrite existing records. To prevent this, validation checks should be used before inserting new data. Unique IDs are always recommended.

Example: Trying to add student with ID 1 again when it already exists.

10. How does FastAPI automatically generate Swagger UI?

FastAPI automatically generates Swagger UI using OpenAPI standards. It reads your route definitions and Pydantic models. This creates interactive API documentation without extra configuration.

Example: Visiting /docs shows all APIs automatically.